Barriers to Designing Ecological Momentary Assessment and Wearable Data Collection Protocols for AI-Driven Substance Use Monitoring in Hawai'i

Yinan Sun[1], Aditi Jaiswal[2], Ali Kargarandehkordi[2], Christopher Slade [2], Roberto M. Benzo[1], Kristina T. Phillips[3], Peter Washington[4]

[1]Division of Cancer Prevention and Control, Department of Internal Medicine, College of Medicine, The Ohio State University Comprehensive Cancer Center, The Ohio State University Wexner Medical Center

[2]Department of Information and Computer Sciences, University of Hawai'i at Mānoa

[3]Center for Integrated Health Care Research, Kaiser Permanente Hawaii

[4]Division of Clinical Informatics and Digital Transformation (DoC-IT), Department of Medicine, University of California, San Francisco (UCSF)

ABSTRACT

Ecological momentary assessment (EMA) and wearable sensors offer unprecedented opportunities to capture the dynamics of substance use through real-time, high-resolution behavioral and physiological data. These data streams are increasingly used to train AI/ML models for digital phenotyping and predictive intervention, raising critical questions about fairness, bias, and inclusivity in model development. However, the adoption of these technologies, or the lack thereof, among diverse and historically marginalized groups raises questions and challenges of equity, cultural relevance, and participant trust. In this study, we conducted a four-week observational study with adults in Hawai'i where we combined continuous Fitbit monitoring and daily EMA surveys to document substance use patterns and cravings. Through semi-structured interviews and grounded theory analysis, we identified six primary barriers to study participation and adherence: (1) disruptions to daily routines, (2) physical and psychosocial discomfort associated with wearing the Fitbit device, (3) concerns about aesthetic compatibility and professional appearance, (4) phone-related issues, (5) challenges related to substance use and cravings, and (6) socially sensitive contexts. We also highlight participant-identified facilitators, such as the value of participant-driven scheduling, motivational feedback, and contextually adaptive protocols. Drawing on these collective findings, we propose a set of design guidelines aimed at advancing the inclusivity, engagement, and fairness of wearable-based EMA research.

1. INTRODUCTION

The Rise of EMA and AI/ML in Substance Use Monitoring
Ecological momentary assessment (EMA) and wearable sensing have become essential tools in digital health and biomedical AI. EMA minimizes recall bias by capturing real-time reports of mood, context, craving, and behavior, while wearables deliver continuous data on physiology, sleep, and movement. When combined with machine learning, these multimodal datasets can identify relapse risk periods and facilitate just-in-time adaptive interventions.

Fairness in Practice: Wearables, EMA, and Inclusion
Despite this potential, real-world use remains limited. Current datasets over-represent higher-income, digitally connected populations, and device design assumptions (e.g., wrist size, skin tone, notification literacy, connectivity) can encode bias before models are trained. As a result, AI/ML systems risk amplifying inequities across groups whose lived experiences differ from those in the training data.

Study Goals
We sought to understand the experiences of participants living in Hawai'i interacting with wearable-linked EMA tools through the context of substance use research as a demonstrative example of a digital health application area disproportionately affecting underserved populations. Specifically, we sought to answer: (1) What sociotechnical and cultural challenges arise when deploying wearable-based EMA tools among a population living in Hawai'i? (2) How can insights from these challenges inform more equitable and culturally responsive approaches to fairness in biomedical AI and digital health research?

METHODOLOGY

Figure 1. User interface of the BanAware mobile app. Participants were asked to complete an EMA 4 times a day at fixed times, asking for self-reported substance use and cravings. Participants had the option of coding substances as fruit names to ensure privacy.

Study Participants
We recruited adults aged 18 years and older residing on O'ahu, Hawai'i through three sites: a harm reduction outreach program, a residential treatment center, and the University of Hawai'i. Eligibility criteria required reliable smartphone access and self-reported active substance use on at least two days per week. Individuals were excluded if they were unable to provide consent, anticipated relocation, or incarceration during the study period.

Device Deployment and EMA Protocol
At the intake visit, participants received a Fitbit Charge 5 to wear continuously for four weeks, which tracked heart rate, step count, sleep quality, breathing rate, and blood oxygen saturation. At the same time, participants completed EMA surveys through the BanAware mobile app.

Data Collection
At baseline, participants completed intake questionnaires assessing sociodemographic characteristics, substance use patterns, and treatment history. Following the four-week monitoring period, participants engaged in semi-structured exit interviews, which lasted 30–60 minutes and were audio-recorded with their consent. These interviews focused on usability, physical comfort, privacy concerns, and sociocultural relevance of the wearable device and mobile application. Field notes were recorded during each session to capture non-verbal cues and contextual observations.

Data Analysis
Interview recordings were transcribed verbatim and analyzed using **Dedoose** software with a **grounded theory approach**. Open coding was done line by line to create descriptive codes, which were then organized into broader themes such as device discomfort, stigma, and emotional stress. In the selective coding phase, these themes were mapped onto the **Social-Ecological Model (SEM)** to capture influences at the individual, interpersonal, community, and societal levels. Two researchers independently coded the transcripts, resolving differences through reflective discussion, memo writing, and iterative comparison with transcripts and field notes to ensure rigor and consensus.

RESULTS

Baseline Characteristics of Participants
A total of 21 participants completed the 4-week wearable-based EMA protocol and interviews. The sample was diverse in gender (male = 11, female = 8, non-binary = 1, transgender = 1), age (23–65 years, mean = 39), and race/ethnicity (White = 6, Native Hawaiian = 5, Samoan = 3, Filipino = 3, Hispanic = 2, Japanese = 1, Chinese = 1). Education ranged from high school (n = 10) to college (n = 6), master's (n = 2), PhD students (n = 2), and one PhD holder. Fifteen participants were employed or enrolled as students, while six were not working. Most (n = 15) logged substances by name, while six used fruit code names for privacy, and seven reported prior experience with wearables.

Sociotechnical Barriers to Wearable-Based EMA Participation

1. Daily Routine: Participants reported difficulty responding to EMA survey prompts during specific periods of the day, such as early mornings, late evenings, work hours, classes, church services, driving, swimming, or while traveling. **(Individual Factors)**
2. Wearable Device Discomfort: Participants reported skin irritation from the Fitbit band, especially when it was wet, and found overnight use uncomfortable or intrusive. Several people preferred not to wear digital devices while sleeping, associating them with a loss of bodily freedom. Continuous use also led to fatigue, reduced relaxation, and feelings of diminished autonomy. **(Individual Factors)**
3. Phone Issues: Participants reported phone-related disruptions, including outdated or slow devices, and switching phones during the study, which led to failing to receive EMA survey prompts or disengagement. **(Individual Factors)**
4. Substance Use and Craving: Participants reported that substance use and cravings interfered with their ability to engage with the study, due to reasons such as irregular sleep patterns, active use periods, and experiencing relapse. **(Individual Factors)**
5. Aesthetic Compatibility and Professional Appearance: Participants expressed concerns related to the aesthetic compatibility of the wearable device with their daily outfits, particularly in social and professional settings. This concern was raised by three female participants and one male participant. **(Individual Factors)**
6. Social Context: participants reported that social environments can be a barrier to EMA compliance. Specifically, they expressed the fear of being judged if others became aware of their substance use or study participation. **(Community and Organizational Factors)**

IMPLEMENTATION GUIDELINES

Drawing on our qualitative findings and participant feedback, we identify design considerations for improving the feasibility, equity, and sustainability of wearable-based EMA studies.

1. Participant-controlled scheduling and context-aware notifications reduce disruption and fatigue.
2. Comfort, customization, and device compatibility are essential for long-term engagement.
3. Digital equity, social adaptability, and cultural relevance can broaden participation and reduce exclusion.
4. Supportive, nonjudgmental re-engagement pathways are critical after life disruptions.
5. Clear definitions and flexible temporal anchoring are essential for accurate event logging.
6. Motivational feedback and tangible progress can sustain engagement beyond initial curiosity.

LIMITATIONS AND FUTURE WORK

This study has several limitations. Eligibility required smartphone access, which might have excluded individuals with unstable housing or limited connectivity. The small, diverse sample limits the ability to generalize findings and perform subgroup analysis, especially among Native Hawaiian and Pacific Islander (NHPI) participants. Our four-week observation period does not capture long-term engagement, and sociotechnical barriers were often linked with broader cultural norms, making it hard to determine causality. Additionally, participants differed in substance type and severity, leading to varied experiences and reducing comparability across groups. Future research should recruit larger, demographically detailed samples, especially among NHPI populations, to identify culturally specific barriers and include Indigenous perspectives in fairness frameworks. Based on these findings, we plan to develop EMA systems that emphasize comfort, customization, and cultural relevance, along with visual feedback and motivational features to maintain engagement. Further studies should examine substance-specific subgroups, situational contexts (such as housing, education, or age), and longer-term deployments to understand how life events and support systems impact participation and outcomes over time.